ANG QI YUEN

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EDUCATION

UNIVERSITI SAINS MALAYSIA Pulau Pinang, Malaysia
Bachelor of Software Engineering with Honours 2026
Electives in Mathematics and Management
CGPA: 3.70/4.00; Dean's List: 5-time holder; Top 10 Dean's List: 1-time holder
Relevant Coursework: Data Structures & Algorithms; Database Design; Software Engineering; Software Testing; Operating Systems; Machine Learning; Linear Algebra & Calculus; Business Intelligence & Analytics

WORK EXPERIENCE

NATIONAL INSTRUMENTS Pulau Pinang, Malaysia
R&D Software Engineer Intern Mar 2025 – Sep 2025

- Built a Python API for device connectivity, using ctypes to call C functions that initiate gRPC communication with devices.
- Enabled Python users to read hardware properties (e.g., voltage, fan status) and send control commands (e.g., adjusting voltage, probing).
- Automated binding generation with Mako templates, removing repetitive hand-written wrapper code.
- Tested the API with pytest as part of the New Product Development (NPD) team, verifying the generated code was functionally correct.
- Migrated the chassis OS build from legacy .mak files to the Yocto Project by rewriting the build logic as Yocto recipes.
- Built and validated a custom Linux image for Raspberry Pi using Yocto.
- Tested embedded hardware using LabVIEW.
- Used GitHub Copilot in day-to-day development and collaborated with the R&D team across software and embedded workstreams.

CRYSTALVIEWHD SDN BHD Pulau Pinang, Malaysia
Trainee Aug 2024 – Jan 2025

- Built and deployed a WordPress website for the APAC IPv6 Council.
- Managed server setup including domain configuration and server settings for hosted services.
- Operated in Linux environments with Vim for cloud system administration and configuration.
- Expanded knowledge of networking and telecommunications, including IPv6, 5G, and SIM card technologies.
- Received cybersecurity training and mentorship under a PhD-qualified cybersecurity specialist.

UNIVERSITY PROJECTS

PROPSYNC
Property Listing Platform – Final Year Project Completed Jun 2026

- Architected a multi-client property listing platform: a Python FastAPI backend on Google Cloud Platform (GCP) serving a Next.js administrative web app on Vercel and a React Native mobile app.
- Built an AI search layer in which Google ADK agents parse natural-language queries (e.g., "3-room condo under 600k") and decompose them into structured sub-queries.
- Embedded listings in PostgreSQL with pgvector for semantic search, matching qualitative attributes such as interior design style that keyword filters cannot capture.
- Constructed nearby points of interest for each property listing, enabling location-aware search and listing details.
- Built a web-scraping data pipeline with Crawl4AI for ingesting, cleaning, and normalising property listings, with scraper runs managed and monitored from the Next.js admin dashboard.
- Trained an XGBoost price-prediction model on pipeline data, with feature engineering and pruning of low-importance features.
- Designed the FastAPI REST API as the single service layer for both clients, using Pydantic models for request and response validation and Supabase Auth for user authentication.
- Modelled the relational schema on Supabase (PostgreSQL), including postcode-to-city normalisation that feeds aggregation tables for area-level comparisons.
- Implemented level-based access control in the admin app with four roles (super admin, admin, moderator, viewer) and an approval workflow for listing management.
- Deployed the backend directly on GCP with OpenStreetMap (OSM) mapping data preloaded, rather than containerising it, avoiding a multi-gigabyte data download on every container initialisation.
- Set up CI/CD pipelines with GitHub Actions to automate build and deployment to Vercel and GCP.

WASTEX
Industrial Waste Marketplace Jan 2025

- Built an industrial waste marketplace with a Next.js frontend and a Django REST backend using the Django ORM.
- Integrated the Google Maps API to provide location-based discovery of nearby waste listings.
- Integrated the GPT-4o API for waste classification and to educate users on the waste they intend to dispose of.
- Delivered the project in a team following Agile/Scrum, with sprint planning and sprint-based delivery.

ACHIEVEMENTS

OCEAN HACKATHON 2025 Brest, France
World Champion Dec 2025

- Developed a Deep Q-Network (DQN) reinforcement learning agent for smart fish feeding, learning feeding policies to reduce feed waste and improve aquaculture efficiency.
- Presented the solution's technical approach and business impact to an international panel of judges.

DERIV AI HACKATHON 2026 Cyberjaya, Malaysia
Semi-Finalist Feb 2026

- Built an agentic AI security pipeline that identifies security issues in websites by combining white-box static analysis, CVE/CVSS vulnerability cross-referencing, and automated black-box testing.

HILTI ITC 2024 Kuala Lumpur, Malaysia
Finalist Jun 2024

- Contributed to a talent acquisition platform built with Next.js and Laravel, focusing on AI-enabled interview features alongside resume screening and candidate rating.

CODENECTION 2024 Kuala Lumpur, Malaysia
Finalist Dec 2024

- Solved timed data structures and algorithms challenges in C++ and Python.

CERTIFICATIONS & PROGRAMMES

AWS CERTIFIED CLOUD PRACTITIONER
Amazon Web Services

- Covered cloud deployment, identity and access management (IAM), and scripting in Bash and Python for cloud operations.

ERICSSON EDGE ACADEMIA PROGRAM
Ericsson

- Completed training in machine learning and AI fundamentals.

LEADERSHIP & ACTIVITIES

GOOGLE DEVELOPER STUDENT CLUB Pulau Pinang, Malaysia
Member, Event Planning and Cultural Development Sep 2022 – Sep 2024

- Organised a TensorFlow talk for 50+ participants covering deep learning concepts and practical applications.
- Organised a Git/GitHub introductory workshop for 200+ participants.

KARATE CLUB Pulau Pinang, Malaysia
Vice President Sep 2024 – Sep 2025

TECHNICAL SKILLS

Languages: Python, C++ (proficient); SQL, JavaScript, TypeScript, Bash, Java (competent).
Backend & APIs: FastAPI, Pydantic, Django (Django ORM), gRPC, ctypes (Python–C interop), RESTful API design, Supabase Auth, third-party API integration (Google Maps, OpenAI).
Databases: PostgreSQL (Supabase, pgvector), MySQL, Oracle SQL, relational schema design.
Testing: pytest, Postman (API testing), LabVIEW (hardware testing).
Cloud & DevOps: Google Cloud Platform, AWS, Vercel, GitHub Actions (CI/CD), Docker, Linux, Git, Yocto Project.
Frontend & Mobile: Next.js, React Native.
AI & Data: XGBoost, Google ADK, semantic search, LLM APIs, Crawl4AI, reinforcement learning (DQN), OpenStreetMap, Power BI, RapidMiner.
AI Coding Tools: Claude Code, OpenAI Codex, GitHub Copilot.
Languages Spoken: Fluent Chinese and English; conversational Malay; elementary Spanish.